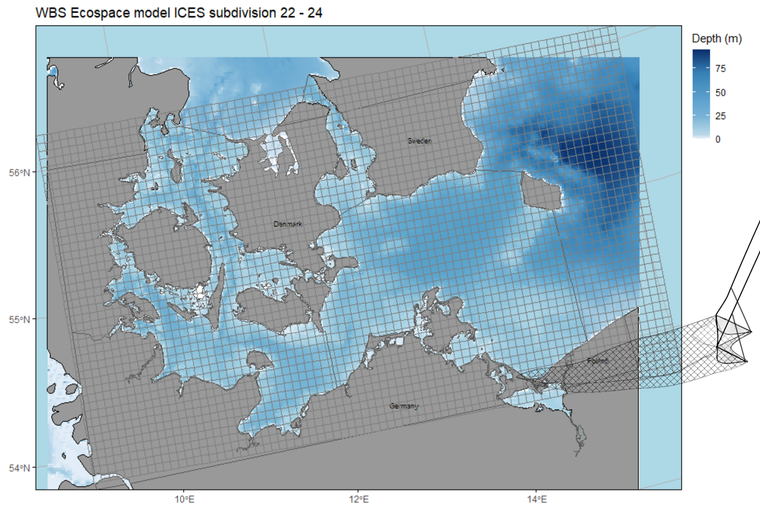
Fisheries and climate shape the future of western Baltic species – a spatially explicit trophic approach

To evaluate how the main commercial stocks could be impacted by climate change and fisheries, the dynamics of the western Baltic Sea ecosystem was modelled using Ecopath with Ecosim and Ecospace, coupled with the outputs from the biogeochemical SCOBI model.



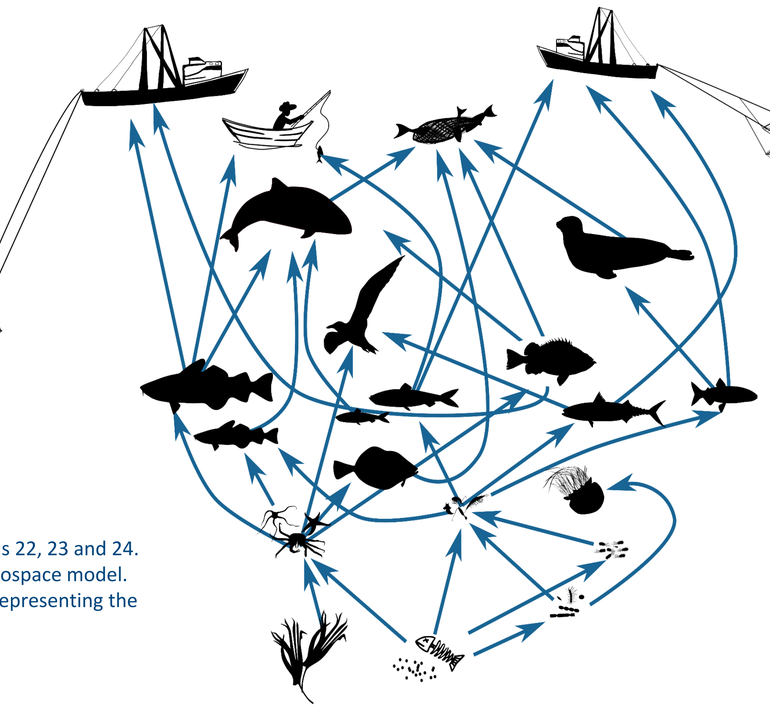


Figure 1: The western Baltic Sea Ecospace model corresponds to ICES subdivisions 22, 23 and 24. The map above shows the overlap between the ecosystem and the grid of the Ecospace model. The food web network on the right illustrates the trophic network, with arrows representing the flow of energy from resources to consumers.

Predicted distribution of the main functional groups was coherent with their observed distribution.

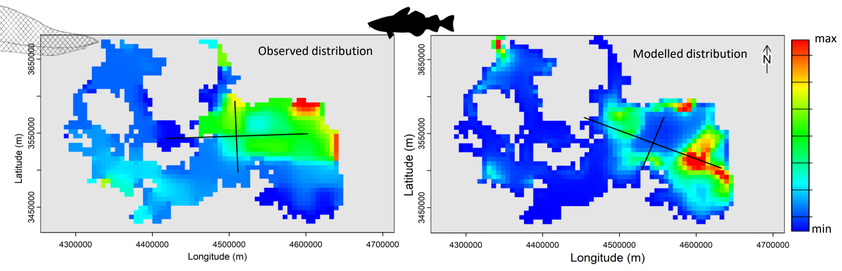
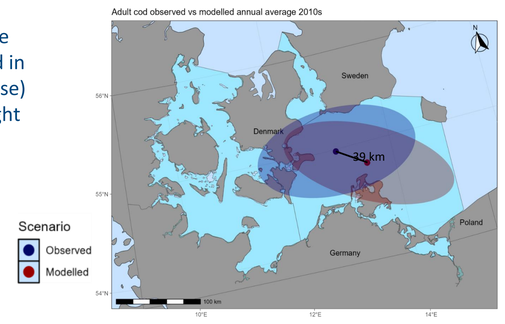


Figure 2: Comparison between the observed distribution of adult cod in the 2010s (left map and blue ellipse) and the modelled distribution (right map and red ellipse).



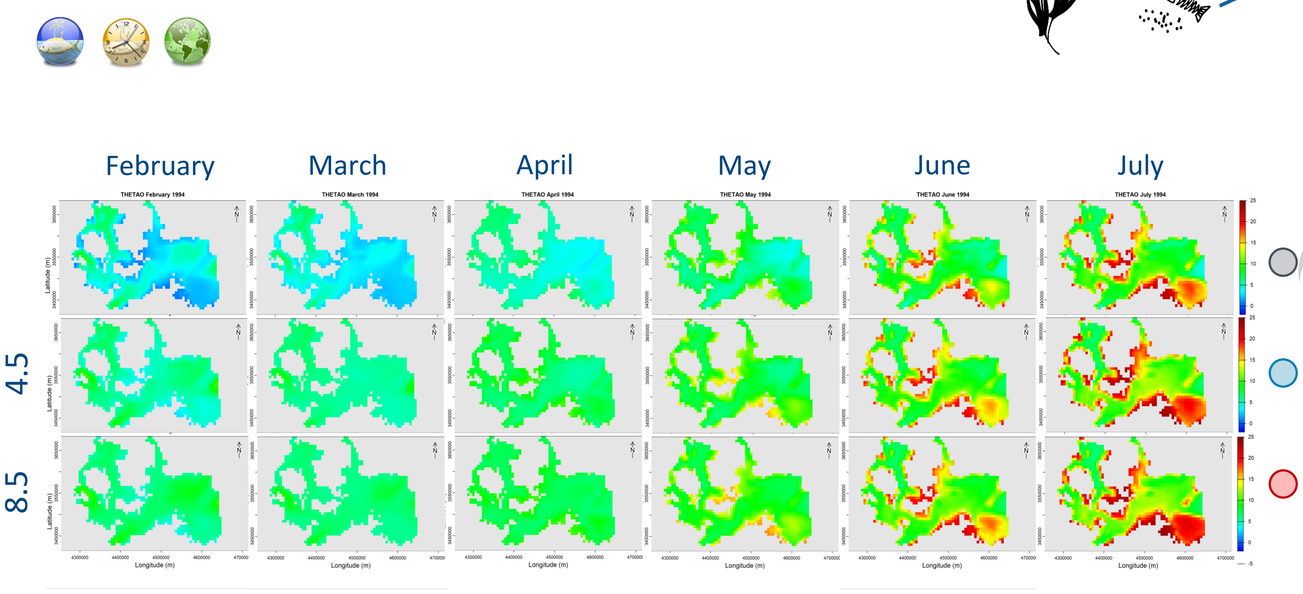


Figure 3: Monthly average of bottom temperature in the western Baltic Sea from the SCOBI model. First row (grey circle) = historical data from 1994; second row (blue circle) = intermediate climate change scenario (RCP 4.5) estimated for 2075; third row (blue circle) = extreme climate change scenario (RCP 8.5) estimated for 2075.

Bottom and surface:

- O_2 concentration
- Salinity
- Temperature

Primary productivity

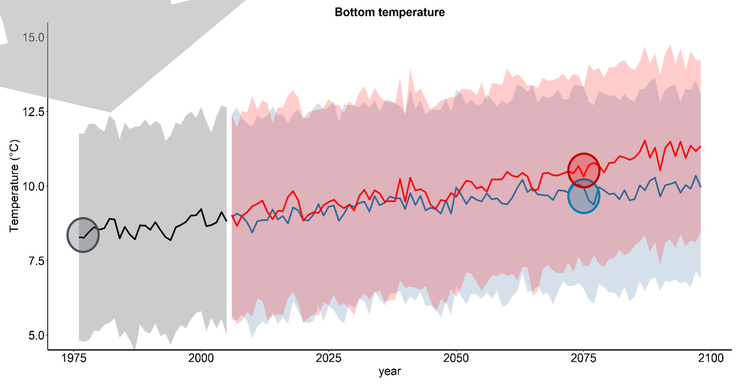


Figure 4: Bottom temperature time series from 1976 to 2005 (grey) and projections bsed on RCP 4.5 (blue) and RCP 8.5 (red) scenarios from the SCOBI model for the western Baltic Sea.

Combined scenarios of climate change and fishing management were simulated

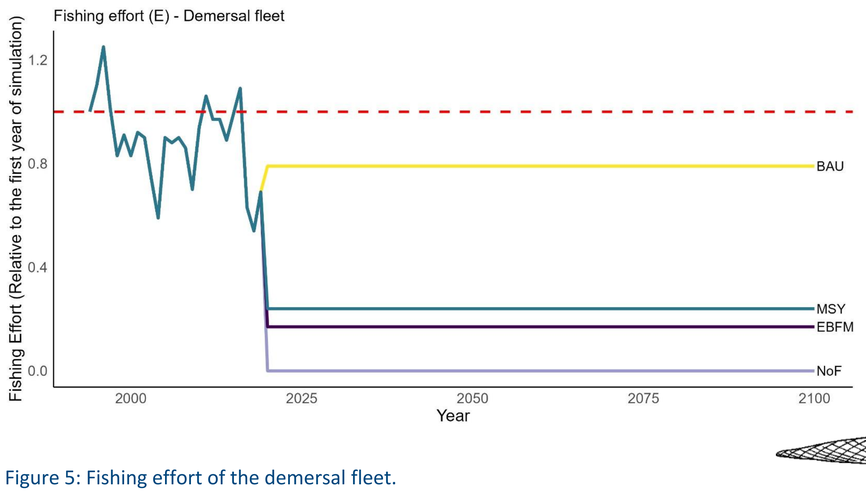


Figure 5: Fishing effort of the demersal fleet.

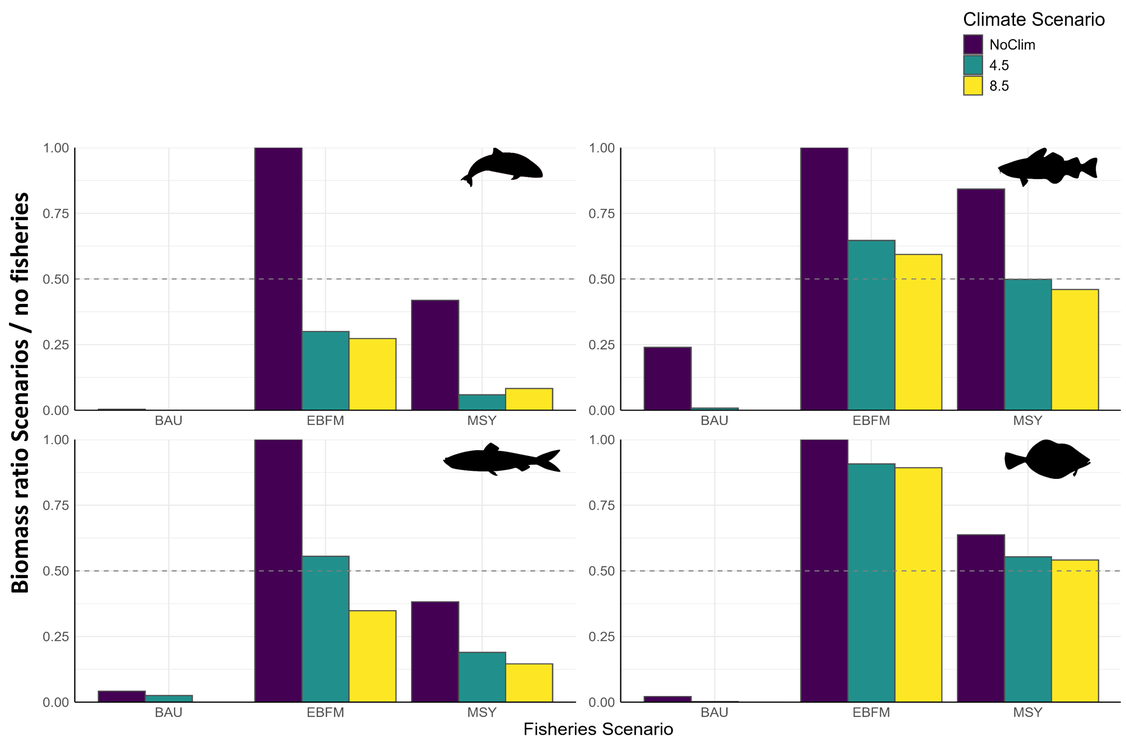


Figure 6: Biomass ratio in 2065 (average of 2055 to 2074) comparing alternative fisheries management scenarios with no fishing. Values above the 0.5 threshold indicate long-term, sustainable management (i.e., $B > B_{MSY}$).

Fishery is the most critical force shaping stock sizes in the western Baltic Sea. The more sustainable strategies (e.g., EBFM) buffer, to some extent, the detrimental effect of climate change



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